

Mutual simulation of Hamiltonian dynamics on n interacting quantum systems

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Control-theoretical model of quantum computers

Time evolution defined by the time-dependent Schrödinger equation

$$\frac{d}{dt}U(t) = -iH(t)U(t), \quad U(0) = \mathbf{1}$$

$H(t)$ Hamiltonian

$U(t)$ is the resulting unitary after time t

$$H(t) = H_d + \sum_{i=1}^m a_i(t)H_i$$

H_d drift Hamiltonian internal to the system

$\sum_{i=1}^m a_i(t)H_i$ control Hamiltonian externally changeable

Complexity of a unitary U : minimal time required to implement U

Fast control limit

strength of the control Hamiltonians arbitrarily large $\implies$

control unitaries arbitrarily fast

Complexity of a unitary U : minimal time

$$U = \exp(-iHt_N)V_N \cdots \exp(-iHt_2)V_2 \exp(-iHt_1)V_1$$

express it as the evolution according to the piecewise constant Hamiltonian

$$U = U_N \exp(-iU_N^\dagger H U_N t_N) \cdots \exp(-iU_2^\dagger H U_2 t_2) \exp(-iU_1^\dagger H U_1 t_1)$$

Average Hamiltonian Theory

Dyson expansion

$$U(t) = \mathcal{T} \exp\left(\int_{\tau=0}^t d\tau H(\tau)\right) = \exp(-i\bar{H}t)$$

Magnus expansion $\bar{H} = \bar{H}^{(0)} + \bar{H}^{(1)} + \dots$

$$\bar{H}^{(0)} = \frac{1}{t} \int_0^t d\tau H(\tau), \quad \|\bar{H}^{(0)}\| \leq 1$$

$$\bar{H}^{(1)} = \frac{-i}{2t} \int_0^t d\tau' \int_0^t \tau'' [H(\tau'), H(\tau'')], \quad \|\bar{H}^{(1)}\| \leq t/2$$

$\implies U(t)$ essentially determined by $\bar{H}^{(0)}$ for small t

Simulating Hamiltonians/First order simulation

H can simulate $\tilde{H}$ with overhead 1, written $\tilde{H} \prec H$, iff

$$\tilde{H} = \sum_{j=1}^N p_j U_j^\dagger H U_j$$

with $\sum_j p_j = 1$ and U_j control operations, i.e.

$\tilde{H}$ is a convex sum of unitary conjugates of H

H can simulate $\tilde{H}$ with overhead τ iff $\tilde{H} \prec \tau H$

time-optimal simulation problem $\longrightarrow$ convex optimization problem

Quantum Networks

system consisting of n nodes $\mathcal{H} = \mathbb{C}^d \otimes \mathbb{C}^d \otimes \dots \otimes \mathbb{C}^d$

pair-interaction Hamiltonian

$$H = \sum_{k < l} \sum_{\alpha \beta} J_{kl; \alpha \beta} \sigma_{\alpha}^k \sigma_{\beta}^l, \quad \sigma_{\alpha} \text{ ONB of } su(d)$$

coupling matrix

$$J = \left(\begin{array}{c|c|c|c|c} 0 & J_{12} & J_{13} & \cdots & J_{1n} \\ \hline J_{21} & 0 & J_{23} & \cdots & J_{2n} \\ \hline \vdots & \vdots & & & \\ \hline J_{n1} & J_{n2} & J_{n3} & & 0 \end{array} \right) \in \mathbb{R}^{(d^2-1)n \times (d^2-1)n}$$

control operations are all local operations in

$$SU(d) \otimes SU(d) \otimes \dots \otimes SU(d)$$

Decoupling/Switching off the time evolution

Conjugating by unitaries of an **error basis** $\mathcal{E} = \{U_1 = \mathbf{1}, U_2, \dots, U_{d^2}\}$

$$\sum_j U_j^\dagger a U_j = \mathbf{0} \quad \text{for all } a \in su(d)$$

cancels all matrices in $su(d)$

Combinatorial problem: which sequences of conjugations allow to switch off all interactions?

Efficient solution: decoupling in networks based on **orthogonal arrays**

number of conjugations grows linearly with n

Action of control operations on the coupling matrix J

Conjugation of the pair-interaction Hamiltonian H by control operations

corresponds to

conjugation of J by a direct sum of orthogonal matrices

$$\begin{aligned} H &\mapsto (u_1 \otimes u_2 \otimes \cdots \otimes u_n)^\dagger H (u_1 \otimes u_2 \otimes \cdots \otimes u_n) \\ J &\mapsto (U_1 \oplus U_2 \oplus \cdots \oplus U_n) J (U_1 \oplus U_2 \oplus \cdots \oplus U_n)^T \end{aligned}$$

Lower bounds on time overhead

If H can simulate $\tilde{H}$ with overhead τ then

$$\tilde{J}/\tau = \sum_{j=1}^N p_j (U_{j1} \oplus U_{j2} \oplus \cdots \oplus U_{jn}) J (U_{j1} \oplus U_{j2} \oplus \cdots \oplus U_{jn})^T$$

convex sum of conjugates of J

$\Rightarrow$ necessary condition given by [majorization criterion](#):

the spectrum $\text{Spec}(\tilde{J})$ majorizes the spectrum $\text{Spec}(\tau J)$

Application I: Time reversal

Time reversal: $H \mapsto -H$ (refocusing, spin echoes)

dipole-dipole coupling between all nodes (easy to invert)

$$H = \sum_{k < l} w_{kl} (3 \sigma_z \otimes \sigma_z - \sum_{\alpha=x,y,z} \sigma_\alpha \otimes \sigma_\alpha), \quad w_{kl} \neq 0$$

lower bound on time overhead is 2

this bound is tight (well known-scheme in NMR)

Application I: Time reversal

strong scalar coupling between all nodes (difficult to invert)

$$H = \sum_{k < l} w_{kl} (\sigma_x^k \sigma_x^l + \sigma_y^k \sigma_y^l + \sigma_z^k \sigma_z^l), \quad w_{kl} \neq 0$$

lower bound on the overhead $n - 1$

the inverted time-evolution is necessarily slower by the factor $(n - 1)$

upper bound $cn - 1$ (decoupling scheme based on orthogonal arrays)

Application II: Hierarchy of Hamiltonians

$$H_1 = \sum_{kl} w_{kl} (\sigma_x^k \otimes \sigma_x^l + \sigma_y^k \otimes \sigma_y^l)$$

$$H_2 = \sum_{kl} w_{kl} (\sigma_x^k \otimes \sigma_x^l - \sigma_y^k \otimes \sigma_y^l)$$

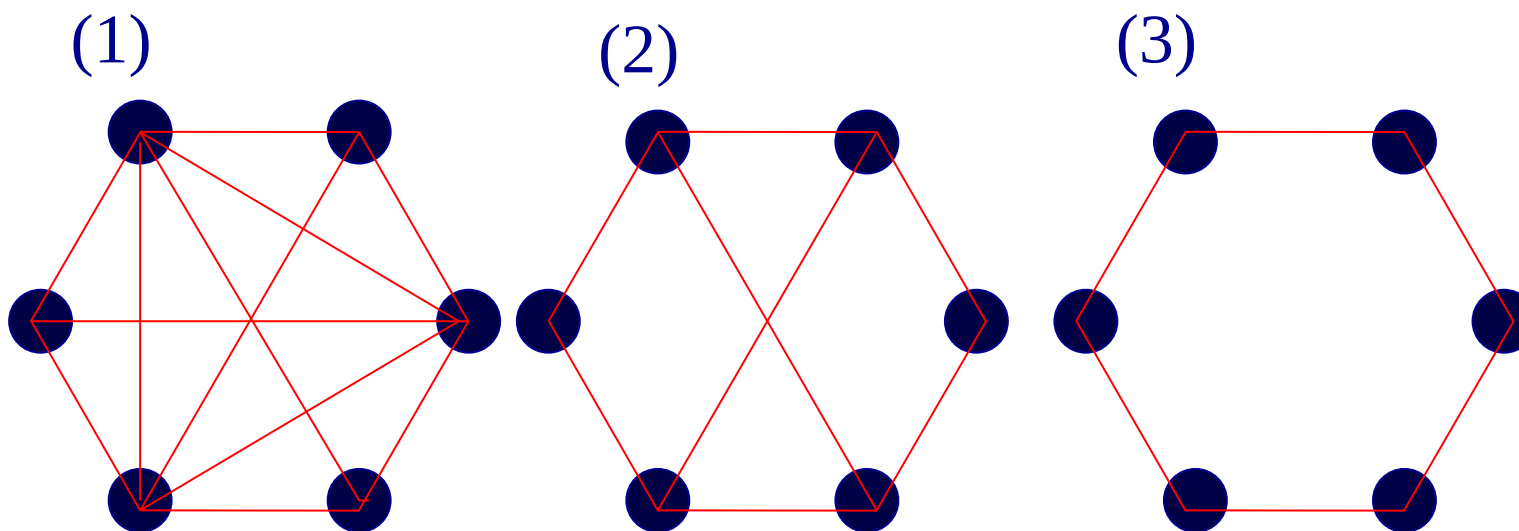
H_1 cannot simulate H_2 with overhead less than $(n - 1)$

H_2 can simulate H_1 with overhead 2

Application III: Switching off **some** interactions

Combinatorial problem: cancel some interactions

Are the other interactions necessarily disturbed?



Yes, for some graphs $\rightarrow$ lower bound on time overhead given by eigenvalues of the adjacency matrix

Upper bound given by chromatic index of interaction graph

Optimal simulation problem equivalent to separability problem

complete zz -Hamiltonian H : interaction between all nodes are of the form

$$\sigma_z \otimes \sigma_z$$

A Hamiltonian with coupling matrix J can be simulated with overhead 1 iff there is a separable n -qubit state ρ such that J is the $3n \times 3n$ -matrix

$$\left(\text{tr}(\rho \sigma_\alpha^k \sigma_l^l) \right)_{kl\alpha\beta}$$

maximal strength of pair-interactions implementable with overhead 1

corresponds to

maximal strength of pair-correlations in a separable state

Universal simulation using finite control groups

Universal simulation does not require all operations in $SU(d)$

Certain subgroups are sufficient; called transformer groups

Finite transformer groups are characterized in terms of group characters

Conclusions

Upper and lower bounds on simulation complexity are given by graph-theoretical notions like chromatic index, chromatic number and graph spectra

Efficient decoupling schemes can be constructed by using orthogonal arrays and error basis

Universal simulation does not require the ability to perform all local unitary operations; we have found finite control groups that are sufficient

Simulating Arbitrary Pair-interactions by a Given Hamiltonian: Graph-theoretical Bounds on the Time Complexity, quant-ph/0106077, to appear in QIC

Simulating Hamiltonians in Quantum Networks: Efficient schemes and Complexity Bounds, quant-ph/0109088, to appear in Phys. Rev. A

Universal Simulation of Hamiltonians Using a Finite Set of Control Operations, quant-ph/0109063, to appear in QIC

Complexity of inverting n -spin interactions: Arrow of time in quantum control, quant-ph/0106085

Lower Bound on the Chromatic Number by Spectra of Weighted Adjacency Matrices, cs.DM/0112023